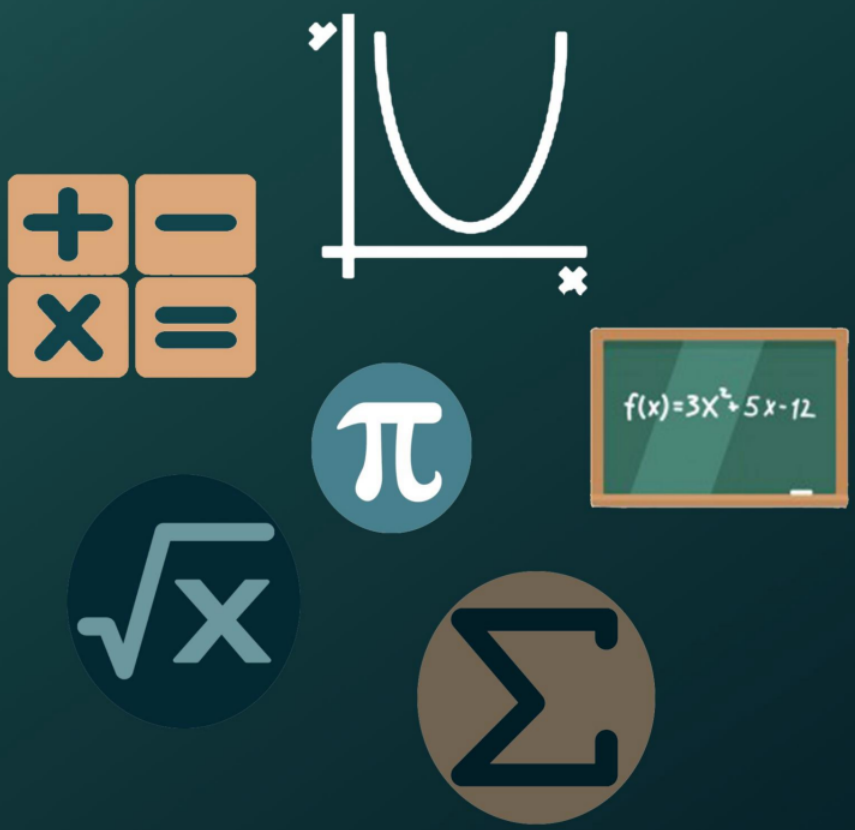
MATHEMATICS

Sets



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-2
1	Introduction to Sets	1	1-2
2	Types of Sets	2-4	3-5
3	Operations on Sets	5-11	6-10
4	Important Results on Sets	12-13	11-13
5	After Complete Chapter		14-23

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Sets

Q1. If A = {2, 4, 5}, B = {7, 8, 9} then n(A × B) is equal to-
(A) 6 (B) 9
(C) 3 (D) 0

Types of Sets

Q2. Which of the following is a null set ?
(A) {0}
(B) {x : x > 0 or x < 0}
(C) {x : $x^2 = 4$ or x = 3}
(D) {x : $x^2 + 1 = 0$, x ∈ R}

Q3. Which set is the subset of all given sets ?
(A) {1, 2, 3, 4, ...}
(B) {1}
(C) {0}
(D) { }

Q4. Which of the following statements is true?
(A) 3 ⊆ {1, 3, 5}
(B) 3 ∈ {1, 3, 5}
(C) {3} ∈ {1, 3, 5}
(D) {3, 5} ∈ {1, 3, 5}

Operation on Sets

Q5. If A and B are two sets, then A ∩ (A ∪ B)' is equal to-
(A) A (B) B
(C) ϕ (D) none of these

Q6. If A is any set, then-
(A) A ∪ A' = ϕ (B) A ∪ A' = U
(C) A ∩ A' = U (D) none of these

Q7. Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}, A={1, 2, 5}, B = {6, 7} then A ∩ B' is-
(A) B' (B) A (C) A' (D) B

Q8. Let A and B be two sets in the universal set. Then A – B equals-
(A) A ∩ B' (B) A' ∩ B
(C) A ∩ B (D) none of these

Q9. Two sets A, B are disjoint iff -
(A) A ∪ B = ϕ (B) A ∩ B ≠ ϕ
(C) A ∩ B = ϕ (D) A – B = A

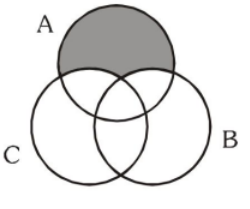
Q10. If A ⊆ B, then A ∩ B is equal to-
(A) A (B) B (C) A' (D) B'

Q11. Sets A and B have 3 and 6 elements respectively. What can be the minimum number of elements in A ∪ B ?
(A) 3 (B) 6 (C) 9 (D) 18

Important Result on Sets

Q12. Let A and B be two sets such that n(A) = 70, n(B) = 60 and n(A ∪ B) = 110. Then n(A ∩ B) is equal to-
(A) 240 (B) 20 (C) 100 (D) 120

Q13. The shaded region in the given figure is-



(A) A ∩ (B ∪ C) (B) A ∪ (B ∩ C)
(C) A ∩ (B – C) (D) A – (B ∪ C)

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If $Q = \left\{x : x = \frac{1}{y}, \text{ where } y \in N\right\}$, then-

(A) $0 \in Q$ (B) $1 \in Q$

(C) $2 \in Q$ (D) $\frac{2}{3} \in Q$

Q2. $A = \{x : x \neq x\}$ represents-

(A) $\{0\}$ (B) $\{\ \}$

(C) $\{1\}$ (D) $\{x\}$

Q3. If $A = \{1, 2, 3, 4, 5\}$, then the number of proper subsets of A is

(A) 120 (B) 30

(C) 31 (4) 32

Q4. Which of the following is a null set ?

(A) $A = \{x : x > 1 \text{ and } x < 1]$

(B) $B = \{x : x + 3 = 3\}$

(C) $C = \{\phi\}$

(D) $D = \{x : x \geq 1 \text{ and } x \leq 1\}$

Q5. $n(A) = n(B) \Rightarrow$

(A) $A \subseteq B$ (B) $B \subseteq A$

(C) $A = B$ (D) none of these

Q6. If A and B are any two sets, then $A \cup (A \cap B)$ is equal to-

(A) A (B) B (C) A' (D) B'

Q7. Let A and B be two sets then $(A \cup B)' \cup (A' \cap B)$ is equal to-

(A) A' (B) A

(C) B (D) B'

Q8. If A and B are two sets, then $A \cup B = A \cap B$ iff-

(A) $A \subseteq B$ (B) $B \subseteq A$

(C) $A = B$ (D) none of these

Q9. If A and B are not disjoint, then $n(A \cup B)$ is equal to-

(A) $n(A) + n(B)$

(B) $n(A) + n(B) - n(A \cap B)$

(C) $n(A) + n(B) + n(A \cap B)$

(D) $n(A).n(B)$

Q10. If $A = \{x : x^2 - 5x + 6 = 0\}$, $B = \{2, 4\}$, $C = \{4, 5\}$, then $A \times (B \cap C)$ is-

(A) $\{(2, 4), (3, 4)\}$

(B) $\{(4, 2), (4, 3)\}$

(C) $\{(2, 4), (3, 4), (4, 4)\}$

(D) $\{(2, 2), (3, 3), (4, 4), (5, 5)\}$

Q11. If A and B are two sets then $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to-

(A) $A \cup B$ (B) $A \cap B$

(C) A (D) B

Q12. In a college of 300 students, every student reads 5 new spapers and every newspaper is read by 60 students. The number of newspapers is-

(A) at least 30 (B) at most 20

(C) exactly 25 (D) none of these

Q13. The set $S : \{1, 2, 3, ..., 12\}$ is to be partitioned into three sets A, B, C of equal size.

Thus $A \cup B \cup C = S$,

$A \cap B = B \cap C = A \cap C = \phi$.

The number of ways to partition S is-

(A) $12!/3!(4!)^3$ (B) $12!/3!(3!)^4$

(C) $12!/(4!)^3$ (D) $12!/(3!)^4$

Q14. If sets A and B are defined as

$A=\{(x,y):y=e^x, x\in R\}$

$B=\{(x,y):y=x, x\in R\}$, then

(A) $B\subset A$ (B) $A\subset B$

(C) $A\cap B=\phi$ (D) $A\cup B=A$

Q15. Let n(A-B)=25+X, n(B-A) = 2X and $n(A\cap B)=2X$. If n(A) = 2(n(B))

then 'X' is -

(A) 4 (B) 5

(C) 6 (D) 7

Q16. Of the members of three athletic teams in a school 21 are in the cricket team, 26 are in the hockey team and 29 are in the football team. Among them, 14 play hockey and cricket, 15 play hockey and foot ball, and 12 play foot ball and cricket. Eight play all the three games. The total number of members in the three athletic teams is

(A) 43 (B) 76

(C) 49 (D) 53

Q17. Let A and B be two sets then

$(A\cup B)^C\cup(A^C\cap B)=$

(A) A^C (B) B^C

(C) ϕ (D) U

Q18. From 50 students taking examinations in Mathematics, Physics and Chemistry, each of the students has passed in at least one of the subject, 37 passed Mathematics, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics, at most 29 Mathematics and Chemistry and at most 20 Physics and Chemistry. Then the largest possible number that could have passed all three examinations is -

(A) 16 (B) 14

(C) 18 (D) 15

Q19. If $P=\{x\in R:f(x)=0\}$

and $Q=\{x\in R:g(x)=0\}$

then $P\cup Q$ is -

(A) $\{x\in R:f(x)+g(x)=0\}$

(B) $\{x\in R:f(x).g(x)=0\}$

(C) $\{x\in R;(f(x))^2+(g(x))^2=0\}$

(D) $\{x\in R;x>1\}$

Q20. Suppose $A_1, A_2, ..., A_{30}$ are thirty sets each with five elements and $B_1, B_2, ..., B_n$ are n sets each with three elements such that $\bigcup_{i=1}^{30}A_i=\bigcup_{j=1}^{n}B_j=S$. If each element of S belongs to exactly ten of the A_i's and exactly 9 of the B_j's, then the value of n is -

(A) 15 (B) 135 (C) 45 (D) 90

Q21. A survey show that in a city that 63% of the citizens like tea where as 76% like coffee. If x% like both tea and coffee, then -

(A) $x=63$ (B) $x=39$

(C) $50\le x\le 63$ (D) $39\le x\le 63$

Q22. An investigator interviewed 100 students to determine their preferences for the three drinks: milk (M), coffee(C) and tea (T). He reported the following : 10 students had all the three drinks M,C,T; 20 had M and C only: 30 had C and T;25 had M and T;12 had M only;5 had C only;8 had T only. Then how many did not take any of the three drinks -

(A) 20 (B) 3 (C) 36 (D) 42

Q23. In a battle 70% of the combatants lost one eye, 80% an ear, 75% an arm, 85% a leg, x% lost all the four limbs the minimum value of x is -

(A) 10 (B) 12 (C) 15 (D) 5

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
B	D	D	B	C	B	B	A	C	A
11	12	13							
B	B	D							

Ex - 2									
1	2	3	4	5	6	7	8	9	10
B	B	C	A	D	A	A	C	B	A
11	12	13	14	15	16	17	18	19	20
A	C	C	C	B	A	A	B	B	C
21	22	23							
D	A	A							